Problem Set 2

Foundations of Machine Learning

- **Submission**: Turn in as a `PDF` and a **separate source code file** on `http://mycourses.unh.edu`. Scanned handwritten notes are encouraged.
- **Questions**: See the class website for a link to office hours

Relevant reading: ILA 1.1–1.3, 2.1, 2.4. Problems marked **[code]** should be submitted as a program in a language of your choice (Julia or Python recommended). The cited problem numbers are from ILA, 5th edition (Strang, 2016).

1. ILA, Problem Set 1.1 (vectors and linear combinations): problems 1, 3, 4, 5, 11, 17, 18, 27, 28.
2. ILA, Problem Set 1.2 (lengths and dot products): problems 1, 2, 4, 10, 14, 32.
3. ILA, Problem Set 1.3 (matrices): problems 1, 4.
4. ILA, Problem Set 2.1 (vectors and linear equations): problems 1, 9, 11, 18.
5. **[code]** Read a CSV data table into a matrix. Compute the dot product of two of its columns, and use a matrix–vector product to normalize every row so that its entries sum to one.
6. **Time spent.** Report approximately how many hours you spent completing this problem set (including the reading).